

Cost function Working mechanism

$$\rightarrow \text{Cost} = \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

$\therefore \frac{1}{2} \|w\|^2$:- is nothing but used to reduce $\|w\|$ so that $\frac{1}{2}$ becomes maximum $\|w\|$

$\rightarrow$ If $\frac{1}{2}$ is maximum that means we have bigger street $\|w\|$ (distance)

Now What is C ?

$\rightarrow$ C is the weight of the penalty

• If a point violates the margin, its ξ contributes to the sum, increasing the objective

• SVM tries to minimize both terms simultaneously

- Make the margin wide

- keep violations small

• C is literally a weight for the second objective

• Larger weight :- SVM prioritizes reducing violations (stricter)

• Smaller weight :- SVM prioritizes wide margin $\rightarrow$ allows more violations

- If C is big - Even a single outlier (point far away) will force SVM to bend a lot to avoid violating it

• This can lead to **overfitting** $\rightarrow$ hyperplane becomes too sensitive to noise

- If C is small : SVM ignores violations a lot $\rightarrow$ hyperplane may misclassify many points $\rightarrow$ **underfitting**

c = Penalty Parameter

Example :- suppose we have 3 points and their violations (ξ_i) are :

Point	ξ_i
1	0
2	0.5
3	1.5

Total Penalty term $c \sum_i \xi_i$

• Case A : Small $c = 1$

- Penalty = $1 \times (0 + 0.5 + 1.5) = 2$

• Case B : Large $c = 10$

Penalty = $10 \times (0 + 0.5 + 1.5) = 20$

Observation :-

- Small $c \rightarrow$ total penalty = 2 $\rightarrow$ violations are "cheap"
(SVM can tolerate them)
- Large $c \rightarrow$ total penalty = 20 $\rightarrow$ violations are "expensive"
(SVM tries to reduce them)

↓

How but ?

- $c = 1 \rightarrow$ wide margin, some points inside allowed
- $c = 10 \rightarrow$ narrower margin, fewer points inside allowed

- Bigger c = penalties are heavy $\rightarrow$ SVM moves margin to avoid violations

- Now, How slack variable (ϵ_i) works

Example

ϵ_i (slack variable)	Meaning
0	Point is perfectly outside the margin, no violation
$0 < \epsilon_i < 1$	Point is inside the margin (margin violated) but still on the correct side $\rightarrow$ marginal violation
$\epsilon_i \geq 1$	Point is on the wrong side of the hyperplane $\rightarrow$ misclassified

ϵ_i measures how much the point violates the margin

$\Rightarrow$ Now for eg let's consider best fit line (hyperplane) :-

$$3x + 2y + 3 = 0$$

- Random points with labels ($y = +1$ or -1)
 $+1$ (positive class)
 -1 (negative class)
- Note: Positive / negative class depends on hyperplane output
- In this example just taking randomly to understand how it works ~~is~~ under the hood.

Point	x	y	Label Layer (y_i)
A	1	1	+1
B	2	1	+1
C	1	3	-1
D	3	2	-1

Note :- E_i (slack variable) cannot be negative therefore it becomes 0

SVM notation :- $w \cdot x + b = 3x + 2y + 3$

Compute slack variables E_i formula :-

$$E_i = \max(0, 1 - y_i (w \cdot x_i + b))$$

- Point A : (1, 1), $y_i = +1$

$$w \cdot x + b = 3(1) + 2(1) + 3 = 8$$

$$E_i = \max(0, 1 - (+1 \times 8)) = \max(0, 1 - 8)$$

$$= \max(0, -7) = 0 \quad (\text{Single } E_i \text{ cannot be negative})$$

(Perfect, outside margin)

- Point B : (2, 1), $y_i = +1$

$$w \cdot x + b = 3(2) + 2(1) + 3 = 11$$

$$E_i = \max(0, 1 - (+1 \times 11)) = \max(0, 1 - 11)$$

$$= \max(0, -10) = 0 \quad (\text{Perfectly, outside margin})$$

- Point C : (1, 3), $y_i = -1$

$$w \cdot x + b = 3(1) + 2(3) + 3 = 12$$

$$E_i = \max(0, 1 - (-1 \times 12)) = \max(0, 1 + 12) = 13$$

(Misclassified)

- Point D : (3, 2), $y_i = -1$

$$w \cdot x + b = 3(3) + 2(2) + 3 = 16$$

$$E_i = \max(0, 1 - (-16)) = \max(0, 1 + 16)$$

$$= \max(0, 17) = 17$$

(Misclassified)

Compute cost function

Soft margin SVM cost function :-

$$\text{Cost} = \frac{1}{2} \|w\|^2 + c \sum_i \epsilon_i$$

Here $w = [3, 2]$

$$\|w\|^2 = 3^2 + 2^2 = 9 + 4 = 13$$

Choose $c = 1$ for simplicity

Total slack sum :

$$\sum_i \epsilon_i = 0 + 0 + 13 + 17 = 30$$

$$\text{Cost} = \frac{1}{2} \times 13 + 1 \times 30$$

$$= 6.5 + 30$$

$$= 36.5$$

∴ This is how SVM calculates ϵ_i for each point, multiplies by c , adds margin cost, and get total cost.

⇒ Imp:- What happens under the hood after getting 36.5

The optimiser (usually a technique like Quadratic Programming) says :

→ "Okay, for the current weights (w, b) , total cost = 36.5. Let's see if we can reduce it.

then it does this again and again

1. Adjust w and b slightly
2. Recomputes cost (maybe 30.2 , then 25.6 , etc)
3. keeps the one that minimizes the cost

so 36.5 is just an intermediate step or final loss value.

↳ its final $\rightarrow$ it means :-

- The SVM has found the best trade-off between having a wide margin and penalizing violations

- so the model has finished training

Imp:- In SVM, the optimiser is a special kind of a mathematical solver that finds the best w and b that minimize the cost

$\Rightarrow$ SVM uses a method called Quadratic Programming (QP)